Muqeeth Mohammed

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Education

- **University of North Carolina - Chapel Hill** Chapel Hill, NC
  Masters of Science in Computer Science *Aug 2021 - May 2023*
  - **Research Interests**: Machine Learning, Natural Language Processing, Transfer Learning, Multitask Learning
- **Indian Institute of Technology Madras** Chennai, India
  Bachelor of Technology (Honors) in Electrical Engineering — GPA:9.36/10.0 (Rank 8/60) *Jul 2016 – May 2020*
  - **Thesis**: Estimation of relatedness between courses in Workflow

Experience

- **University of North Carolina - Chapel Hill** Chapel Hill, NC
  Research Assistant — Advisor: Prof. Colin Raffel *Aug 2021 - present*
  - Identified issues in learning effective routing strategies using gradient estimation methods to train models with conditional computation.
  - Working on designing better techniques to route in models with a mixture of experts that learn multiple tasks.
  - Worked on showing few-shot parameter efficient fine-tuning is better in cost and performance than in-context learning.
- **Indian Institute of Technology, Delhi** Delhi, India
  Research Assistant — Advisors: Prof. Mausam, Prof. Soumen Chakrabarti *Jul 2020 - Jul 2021*
  - Identified inconsistencies when translating OpenIE data from English to other languages. We built a system that overcame these inconsistencies and provided state-of-the-art on the Open IE task over five languages.
  - Created and publicly released OpenIE test sets for low-resource Indian languages like Hindi and Telugu.
  - Worked on a unified labeling architecture for structured prediction tasks that include syntax, semantics, information extraction, and sentiments in NLP which is 3x faster than state-of-the-art achieving nearly the same accuracy.
- **Senior Thesis** Chennai, India
  Undergrad Student — Advisors: Prof. Sutanu Chakraborti, Prof. Anil Prabhakar *Jan 2020 - May 2020*
  - Designed a retrieval system to retrieve relevant courses to a given query course from our university courses.
  - Used explicit semantic analysis with the addition of phrases to achieve 0.868 nDCG score compared to a neural model baseline of 0.735. Implemented the idea of retrofitting to user feedback to further improve the score to 0.92.
- **Flipkart** Bengaluru, India
  Summer Intern *May 2019 - July 2019*
  - Built an unsupervised classification model to classify opinion-aspect pairs from Flipkart review corpus for shoes into categories. Achieved an accuracy of 76 % on human-annotated test set.
  - Implemented decision trees using features from the review sentence's dependency parse to identify the categories like Quality, Comfort, etc. implicitly or explicitly spoken in a review with an F1 score close to 86% for each category.

Publications

- **Models with Conditional Computation Learn Suboptimal Solutions (ICBINB@NeurIPS, 2022)**
  ***Mohammed Muqeeth**, Haokun Liu, Colin Raffel*
- **Few-Shot Parameter-Efficient Fine-Tuning is Better and Cheaper than In-Context Learning (NeurIPS, 2022)**
  Haokun Liu, Derek Tam*, **Mohammed Muqeeth***, Jay Mohta, Tenghao Huang, Mohit Bansal, Colin Raffel*
- **Alignment-Augmented Consistent Translation for Multilingual Open Information Extraction (ACL, 2022)**
  Keshav Kolluru, **Mohammed Muqeeth***, Shubham Mittal, Soumen Chakrabarti, Mausam*

MISCELLANEOUS

- Received Masters Merit Fellowship which covers full tution and provides stipend for the first year at UNC Chapel Hill
- Recevied RamShriram Merit Scholarship at IIT Madras for maintaining 8.0+ GPA every year
- Reviewer: ACL 2022, EMNLP 2022

Technologies

- Python, PyTorch, Sklearn, Numpy, LaTeX, bash, git